

Chapter 6

The Power Supply

The basic function of power supply is to convert the type of electrical power available (Alternate Current-AC) to the type computer circuitry can use (Direct Current). Power supply is designed to convert 120v (American standard) or 220v (European standard) Alternating Current (AC) into +3.3v, +5v, and +12 Direct Current (DC) needed by the computer components. Usually the digital electronic components and circuits in the system (motherboard, adapter cards, drive logic boards) use +3.3v or +5v power. The motors like disk drive motors, and fans use +12v power.

Fact: Almost everything in your home runs off standard AC power. But they actually use

DC internally. E.g printers, external modem, scanner, etc. This necessitates conversion of the AC power to DC power.

In addition to supplying power to the system, the power supply also ensures that the system does not run unless power supply is sufficient to operate the system properly. In other words, the power supply actually prevents the computer from running (starting up) or operating until all correct power levels are present. Each power supply completes internal checks and tests before allowing the system to start. If tests are successful, the power supply sends a special signal called Power_Good to the motherboard. If this signal not present, computer does not run.

Types of Power Supplies

Power supplies are available in a few standardized sizes and shapes. There are two classes of power supply: obsolete and modern

Obsolete:

- ✓ PC/Xt style
- ✓ AT style

Modern:

- ✓ LPX style
- ✓ ATX style
- ✓ SFX style

AT Power Supply

A pair of almost identical connectors designated P8 and P9 links the power supply to the motherboard. Each of these connectors have 6 pins. When you plug these connectors to the motherboard, keep the black connectors together i.e. both in the middle unless stated on the motherboard manual.

The following table shows the voltage values for each color coded wires on P8 and P9

<u>Cable Color</u>	<u>Voltage</u>	
<u>tolerance</u>		
Yellow	+12	±10%
Blue	-12	±10%

Red	+5	±5%
White	-5	±5%
Black	0(ground)	N/A

<u>P8</u>	<u>P9</u>
Silver O power good	black O ground
Red O +5V	black O ground
Yellow O +12V	white O -5V
Blue O -12V	red O +5V
Black O ground	red O +5V
Black O ground	red O +5V

Caution: plugging the power connectors in the wrong direction could damage the computer Pay attention when you plug.

ATX Power Supply

ATX uses a single P1 connector instead of P8 and P9 connectors. This has an advantage of not plugging in the wrong direction which is a common problem in AT power supplies. P1 connector include 3.3V wire along with the standard 5V and 12V wires. It has 20 pins on the connector.

<u>Cable Color</u>	<u>Voltage</u>
Yellow	+12V
Red	+5V
Blue	-12
Black	0(ground)
Brown	+3.3V
Magenta	+5V stand By
Silver	Power Good
Green	+5V remote power on/off
White	-5V

ATX power supply supports soft power feature which means the operating system (software) can turn off the PC. It also support system to cool itself.

Remark: the power supply generates not onl +3.3V, +5V, and +12V but also -5V, and -12V. The positive voltages gives power to everything in the system. What are the negative voltages used for? The answer is not much. Some communication circuits have used -12V power as a bias current but rarely is -5V used at all today. The newst power designs no longer include -5V. -5V is required on ISA bus.

LPX Style

The next power supply unit to gain popularity was the LPX style. It is also called PS/2, or slim type. The LPX power supply has exact same motherboard and disk drive connectors as AT standard; it differs mainly in the shape. LPX systems were designed to have smaller foot print and lower height than AT-sized systems. LPX systems

needed smaller power supply. The PSU designed for LPX systems is smaller than Baby AT style in every dimension and takes up less than half of the space of its predecessors.

NLX Style

The NLX specification is developed by Intel. It defines a low profile case and motherboard with many of the same attributes as ATX. For interchangeability, NLX systems were designed to use ATX power supply even though the case and motherboard dimensions might be different.

SFX Style

Intel recently released a third new motherboard specification called Micro-ATX. This motherboard is intended for low end systems which are designed with an even smaller foot print than NLX, and with potentially smaller power supply requirements. It uses very small power supply called SFX (Small Form Factor). It can also use ATX power supply. SFX is specifically designed for use in small systems containing a limited amount of hardware. SFX is wholly a separate standard and wholly compliant with other motherboards as well. SFX uses the 20-pin connector defined in ATX standard.

ATX Optional Connector

In addition to the 220 pin power connector, the ATX specification defines an optional 6 pin connector (two rows of three pins each). The computer uses these signals to control and monitor the cooling fan, monitor 3.3V signal to the motherboard and provide power and grounding to devices. It is called P2.

Connectors to Peripheral Hardware

Two standard types of connectors are used to connect power to peripheral devices:

- ✓ Molex connectors
- ✓ Mini connectors

Molex Connector

This is the most commonly used connector. It provides both +5V and +12V power. Hard disk drives, internal tape drives, CD-ROM drives, DVD drives, and older 5^{1/4} floppy drives use this power connector. It has four pins.

<u>Pin</u>	<u>wire color</u>	<u>signal</u>
1	yellow	+12V
2	black	ground
3	black	ground
4	red	+5V

Mini Connector

It is used primarily for 3.5 floppy disk drive. It has four pins and four wires. Connecting the connector backward could damage the system. The good thing is it does not fit in the wrong direction

Power Failure

Power supplies are affected by quality of local power source. Though we don't notice sudden changes in the quality of electrical power, computer and other sensitive electronics are affected. Common power delivery problems are:

- ✓ Spikes
- ✓ Surges
- ✓ Sags
- ✓ Brown outs black outs

Surge

This is a brief increase in the voltage source. This is sometimes catastrophic. This may originate from power source but most often due to lightning strike.

Spikes

Spikes are very short over voltage conditions. Spikes are measured in nanoseconds, whereas surge is measured in milliseconds.

Sags

A brief decrease of voltage at power source.

Brown Outs

If a sag lasts longer than 1 second, it is called brown out. The overloading of primary power source can cause this.

Black Outs

This is a complete power failure. This could be caused by equipment failure, or accidental cutting of power cable. When a power returns after black out, there is power spike and danger of a power surge.

Power Control

There are different devices that enable us to control power. Their function varies from device to device depending on what we use.

a) UPS(Uninterruptible Power Supply):

A UPS helps in black out situations when the power is gone totally, and brown outs, when voltage is low though there is power. UPS has batteries that provides back up power. When the power is gone, the battery provides voltage to the PC so that you can save your work. But it does not provide unlimited power. There are two types of UPS:

Standby UPS:

SPS has a battery that begins generating power as soon as the unit detects a sag in the power supply. It takes only a split of seconds to come online. The disadvantage is that till the UPS become online, your data might get lost.

Online UPS:

It provides electricity to the PC all the time. It uses electricity from the AC outlet to simply recharge its batteries. When the power goes, the data is not affected because the UPS is supplying power.

b) Surge Suppressor:

Surge suppressors help to absorb power surges so that your computer does not feel their effects. They come as either separate modules or incorporated in the UPS.

c) Voltage Stabilizer

It is a transformer that deliver relatively constant output when output voltage changes over time. The output voltage is regulated using transistor.

Battery

Computer has a Chip that combines real time clock and non-volatile memory. This chip is the CMOS chip(CMOS RAM). They are designed to consume low power.

The CMOS chip stores basic system configuration like amount of memory installed, type of floppy and hard disk drives, type of keyboard and mouse, etc. This is powered by battery when system is off. This battery keeps the time chip running and the CMOS powered. Most systems use lithium type battery because they have very long life.

Most of the higher quality systems sold today have a new type of chip that has the battery embedded in it. These chips are notable for long life. Under normal conditions, the battery lasts for 10 years. Most of the time, these chips-battery combinations are installed in a socket on the motherboard. Some systems do not use battery at all.